

A Packing of 17 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.549645. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.540835, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.549645 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 17$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.540835, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 17/16 = 2.550000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N17.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
115892969413809891680282931487394679315899362619742343426330528115230124
183577782190584358208944444649944400472566111569133087450029889619565719
512227925606360703673551170497341571722221044264
```

denominator:

```
45454545454443777449152363188184212210392430928553792174503597905321566
755924532126788092769755712602720794509936966079834181196578685591708336
16924005622688148089508296327746053591356111075
```

B Coordinates

17 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N17.txt`.

```
230281729452 -27731676942 418113270331 488614564039398387704765 -396165171160 602109872513 6130224313677657970369 -7206189642 -114616819585 41162133058
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